

Enhancing Causal Structure Learning: Variance Reduction and Adaptive Cutoff Selection for the NoTEARS Algorithm

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<https://github.com/waytlion/MathML-internship-2025>

Abstract—The NoTEARS algorithm has emerged as a landmark approach for continuous optimization in causal structure learning. However, its performance is highly sensitive to the signal-to-noise ratio and the choice of the hard-thresholding cutoff parameter. In this work, we propose two primary enhancements to increase the robustness of NoTEARS in challenging data regimes. First, we implement a bootstrap aggregation framework to reduce model variance, demonstrating significant improvements in Structural Hamming Distance (SHD) for datasets with weak causal signals. Second, we introduce a novel, data-driven approach for selecting the cutoff parameter via cross-validation using a modified Mean Squared Error (MSE) metric that incorporates an L_0 penalty. Our experiments on synthetic and benchmark datasets show that these modifications allow for near-optimal graph recovery in scenarios where standard fixed-cutoff configurations fail, particularly when causal relationships are stable but weak.

I. INTRODUCTION

This report summarizes the results of our research project conducted during the Math & Machine Learning Internship. Our goal was to deeply understand and improve the performance of the NoTEARS algorithm for causal structure learning [1]. To this end, we investigated how weight intervals affect its accuracy, applied bagging to reduce variance in noisy settings, and proposed a cross-validation approach for selecting the cutoff parameter. The following sections present our methodology, experiments, and key findings.

II. INVESTIGATING THE EFFECT OF WEIGHT INTERVALS

In the first part of our work, we ran an experiment to test our understanding of the NoTEARS algorithm and its implementation. For this purpose, we formulated the following hypothesis.

Smaller weight initializations during random graph generation will lead to worse performance of the NoTEARS algorithm, because the impact of the linear signal becomes less dominant compared to the additive noise term (given the noise remains unchanged).

The NoTEARS algorithm is computed on data generated from randomly generated DAGs. In the standard setup, DAGs are initialized with k edges, and edge weights are assigned by drawing uniformly from given weight intervals. In the original paper, experiments were run on DAGs where weights were assigned from intervals $[-2, -0.5]$ and $[0.5, 2]$.

To investigate the effect of smaller weights, we tested 10 smaller weight intervals on randomly generated DAGs with 20 nodes and 20 edges each. To compare the estimated DAGs produced by NoTEARS to the ground truth graphs, we used the SHD (Structural Hamming Distance) metric. SHD measures the number of edge additions, deletions, or reversals needed to transform the estimated graph into the ground truth, providing an intuitive and summarizing way to assess graph recovery.

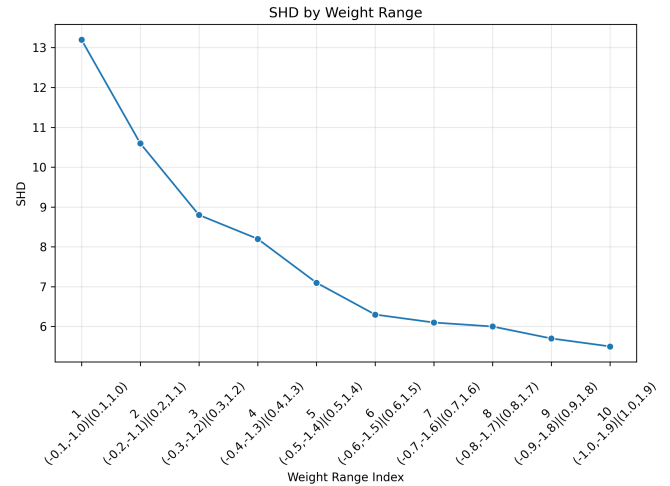


Fig. 1. Effect of decreasing weight ranges on SHD. The x-axis shows the different weight ranges; the y-axis shows the corresponding SHD values.

As shown in Figure 1, our hypothesis was validated by the experiment: smaller weight ranges led to consistently higher SHD, indicating worse recovery of the true graph structure.

III. BAGGING AND VARIANCE REDUCTION

After observing that the performance of the algorithm worsened for DAGs with smaller edge weights, we investigated how to mitigate this effect. As previously discussed, reducing the range of edge weights lowers the strength of the linear signal in the data. Since our model is based on linear effects (via a linear SEM), smaller weights make the linear signal less dominant compared to the additive Gaussian noise.

This leads to datasets with a lower signal-to-noise ratio, which makes recovering the underlying causal structure more difficult. In statistical learning theory, the prediction error of a method can be decomposed into bias, variance, and irreducible error. Here, the variance component reflects the model’s sensitivity to fluctuations in the training data — especially due to noise.

When the signal-to-noise ratio becomes lower, the model is more affected by noise, leading to higher variance and, consequently, a higher prediction error. To address this, we explored methods to reduce variance and were led to bagging, which we implemented.

Bootstrap aggregation, or *bagging*, is a general-purpose method to reduce the variance of statistical learning procedures [2]. High variance means that splitting the training data randomly into two parts and fitting the same model could yield very different results.

To understand how bagging reduces variance and leads to better results, we consider the statistical property of the sample mean. Given a set of n independent observations $Z_1, Z_2, \dots, Z_n$, each with variance σ^2 , the variance of their mean $\bar{Z}$ is given by:

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i$$

Using the properties of variance yields the following:

$$\text{Var}(\bar{Z}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n Z_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Z_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

This shows that averaging independent estimates reduces the variance by a factor of n . Bagging applies this principle by averaging predictions over multiple bootstrapped samples and models, thereby stabilizing the predictions and reducing overall variance.

A. Bagging Implementation and Results

We implemented bagging in the same experimental setup to test whether it could improve the NoTEARS algorithm’s performance. Again, we used SHD as the comparison metric.

To ensure robustness against randomness (e.g., that all weights happen to come from the lower end of the interval), we repeated the experiment with 10 different random seeds. In each seed round, we generated 20 different bootstrap samples from the data and fitted the linear SEM on each sample. We then aggregated the estimated weight matrices by averaging them.

As shown in Figure 2, bagging indeed improved the algorithm by lowering the SHD across all weight intervals. The improvement is especially notable for smaller weight ranges, where variance has an increasing positive effect.

B. Bagging Results on Benchmark Dataset

In addition to testing bagging on randomly generated graphs, we also applied our bagging approach to the real-world benchmark dataset used in Section 5.4 of [1]. This

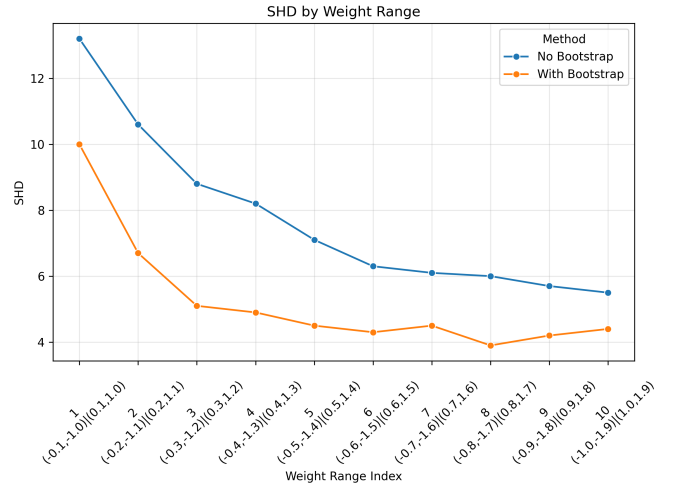


Fig. 2. Effect of bagging on SHD across weight intervals.

dataset consists of $n = 7466$ samples, $d = 11$ variables, and approximately 20 edges.

Interestingly, unlike in our synthetic experiments, applying bagging to this benchmark dataset did not lead to any noticeable improvement in performance, as measured by the SHD metric. We hypothesized that this might be due to the large number of samples in the benchmark dataset. With so much data, the variance of the model’s estimates is already quite low, so bagging cannot meaningfully reduce it further.

To investigate this further, we repeated the experiment using only $n = 100$ samples from the same benchmark dataset—matching the sample size we used in our synthetic graph experiments. Indeed, with this smaller sample size, our bagging approach reduced the SHD from 22 to 20.

IV. INVESTIGATING THE CUTOFF PARAMETER

Another aspect that came to our attention when reviewing the paper was the cutoff parameter. Its original purpose is to trim the resulting matrix of coefficients so that small weights are set to zero and the acyclicity of the resulting DAG is enforced. The authors use a constant cutoff parameter of 0.3 for all their experiments, which is a good choice for the generated weights that are uniformly sampled from the union of the intervals $[0.5, 2]$ and $[-2, -0.5]$.

When we generate random Bayesian Networks with weights sampled from smaller ranges, e.g. $[0.1, 1]$, this cutoff is clearly not optimal. If the actual parameter takes a value smaller than 0.3, it is cut off even when the model learns the causal relationship perfectly.

Since the authors themselves suggest using a suitable data-driven choice for the cutoff parameter, we investigated this problem further. For artificially generated DAGs, one could choose the cutoff to match the weight interval, but this interval is usually not known in advance for real-world datasets.

One approach that came to mind was cross-validation as a model-agnostic way of finding optimal hyperparameter configurations. We implemented a simple 5-fold cross-validation on

the generated data and used the mean squared reconstruction error as a metric:

$$\text{MSE} = \frac{1}{n \cdot d} \sum_{i=1}^n \sum_{j=1}^d (X_{ij} - \hat{X}_{ij})^2$$

where the reconstructed matrix is defined by $\hat{X} = XW$, with $W_{jj} = 0$ for all j .

The reconstruction error often increased monotonically with larger cutoff values, which implied the best-performing model was the one with no cutoff at all. But when comparing the graphs learned at different cutoff values to the known ground truth graph, the optimal cutoff was significantly greater than zero in every configuration.

This problem most likely arises because the algorithm already uses an L_1 penalty term to train sparse DAGs, so removing learned parameters via a cutoff simply increases MSE. We also investigated the true and false positive rates of the different graphs, which showed a positive effect of a well-chosen cutoff. The false positive rate decreased rapidly with larger cutoffs, while the true positive rate only decreased slowly.

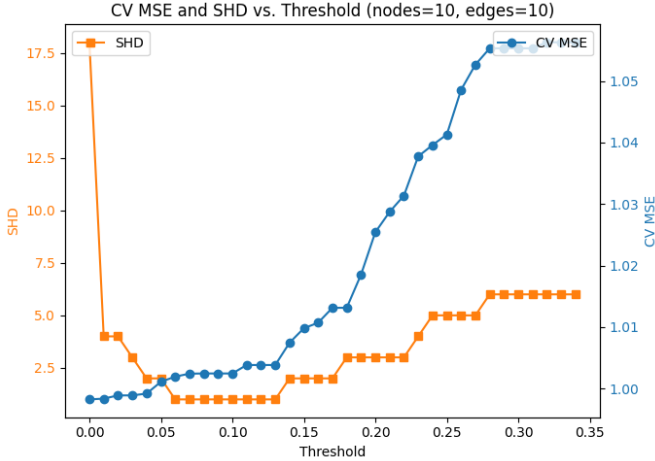


Fig. 3. Example of reconstruction MSE of a learned graph for different cutoff values. Note that the MSE increases monotonically.

We therefore modified the scoring metric used in cross-validation to account for this problem. Since models learned at cutoffs close to zero exhibited a very high false positive rate, we introduced the number of non-zero entries in the learned adjacency matrix as a penalty term. To enable comparison across different configurations, we normalized by the number of nodes d , which is known in advance even for real datasets. This resulted in the modified squared reconstruction error:

$$\text{Modified MSE} = \text{MSE} + \alpha \cdot \frac{\|\hat{W}\|_0}{d}$$

where $\|\hat{W}\|_0$ denotes the number of non-zero edges and d is the number of nodes.

A. Cross-Validation Experiments for Cutoff Selection

We evaluated different values for the penalty scalar α on randomly generated graphs with 10 nodes and edges, and 100 datapoints, with weights sampled from the union of $[-1, -0.1]$ and $[0.1, 1]$. A scalar value of $\alpha = 0.01$ showed good results, with the cross-validated cutoff matching the SHD-optimal cutoff. The additive noise was kept constant at $\varepsilon \sim \mathcal{N}(0, 1)$.

We simulated DAGs with different numbers of nodes and edges as well as different weight range configurations. The number of datapoints was always set to ten times the number of edges. Figure 4 shows the results for configurations where the number of nodes and edges was equal.

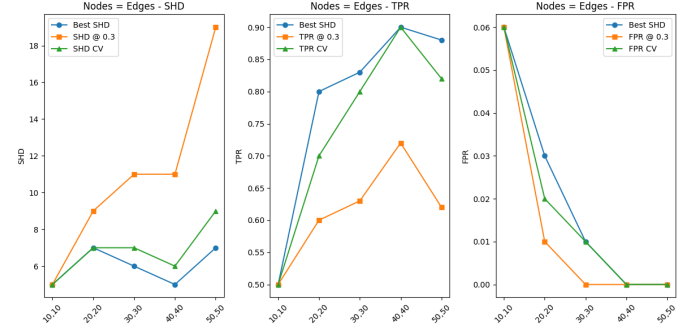


Fig. 4. SHD, TPR, and FPR of DAGs learned with optimal, cross-validated, and vanilla cutoff. The x-axis shows the number of nodes/edges in the randomly generated graphs.

We evaluated the method for node-to-edge ratios $r \in \{0.5, 1, 1.5, 2, 3\}$. The cross-validated cutoff resulted in a lower SHD than the vanilla cutoff of 0.3 in almost all experiments. The true positive rate was often much better using the tuned parameter, while the false positive rate was only marginally higher.

We must keep in mind that there is a large class imbalance, since the number of false edges the model can learn is much higher than the number of true edges. Nevertheless, we achieved a good way to find a near-optimal cutoff parameter regarding the SHD of the learned graph. With the ratio of edges to nodes increasing, the performance worsened slightly but remained very stable. It happened rarely that the SHD was worse than at 0.3, and if so only slightly (see Appendix A).

When we repeated the experiments with weights from $[-0.5, -0.05]$ and $[0.05, 0.5]$, the results were also good, while the average SHD was larger due to the lower signal-to-noise ratio. However, for the weight configuration used in the original paper ($[-2, -0.5]$ and $[0.5, 2]$), the cross-validated cutoff did not consistently outperform the vanilla cutoff. It was often chosen too low, while a cutoff of 0.3 frequently achieved an SHD of 0.

B. Discussion

The cross-validated cutoff using the modified reconstruction error showed strong results for datasets where causal relationships are stable but weak. For datasets where noise is small relative to the signal, it can lead to worse results, so using

this method unconditionally is not recommended. However, in combination with uncertainty estimation for the parameters, or methods to estimate the signal-to-noise ratio, the cross-validated cutoff could achieve better true positive rates than a hand-picked constant parameter.

V. CONCLUSION

In this internship project, we investigated three aspects of the NoTEARS algorithm for causal structure learning and implemented practical improvements for challenging scenarios.

We first validated that smaller edge weight intervals lead to worse performance due to reduced signal-to-noise ratios. We then implemented bootstrap aggregation (bagging) as a variance reduction technique, which consistently improved graph recovery accuracy, especially for smaller weight ranges and limited sample sizes. Finally, we developed a cross-validation approach for automatically selecting the cutoff parameter using a modified reconstruction error metric that penalizes false positive edges.

Our bagging approach showed clear benefits in low signal-to-noise scenarios but had minimal impact on large datasets where variance is already low. The cross-validation method for cutoff selection worked well for weak-signal cases, achieving near-optimal performance and improving true positive rates. However, it was less effective in high signal-to-noise scenarios where the original fixed cutoff of 0.3 performed well.

These findings suggest that enhancement strategies should be adapted based on data characteristics. The bagging method offers a straightforward way to improve robustness in small-sample or noisy conditions, while the adaptive cutoff selection can be valuable when dealing with weak causal relationships.

REFERENCES

- [1] X. Zheng, B. Aragam, P. Ravikumar, and E. P. Xing, "Dags with no tears: Continuous optimization for structure learning," 2018. [Online]. Available: <https://arxiv.org/abs/1803.01422>
- [2] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning: with Applications in R*, 1st ed., ser. Springer Texts in Statistics. Springer, 2013.

APPENDIX

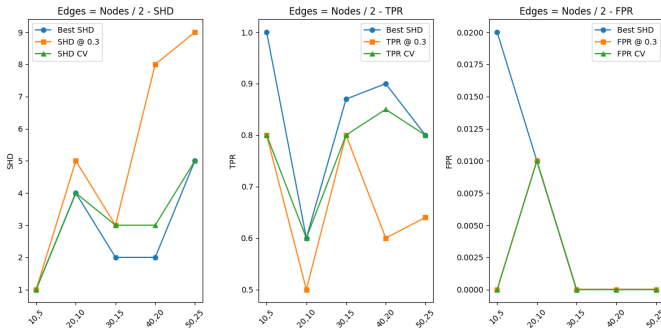


Fig. 5. Results for factor 0.5 configuration.

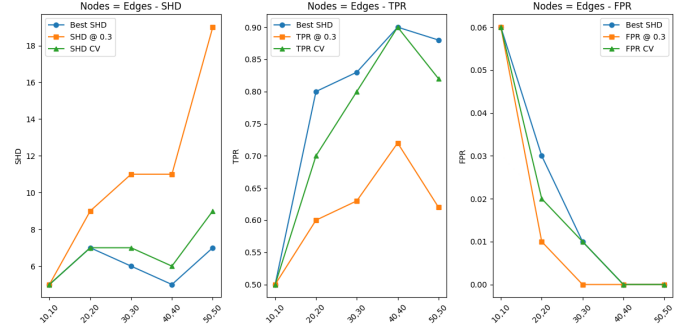


Fig. 6. Results for factor 1 configuration.

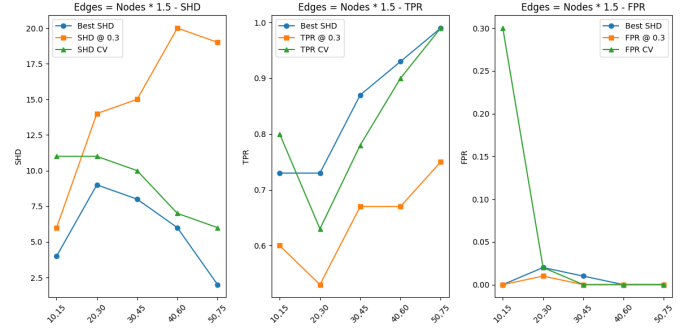


Fig. 7. Results for factor 1.5 configuration.

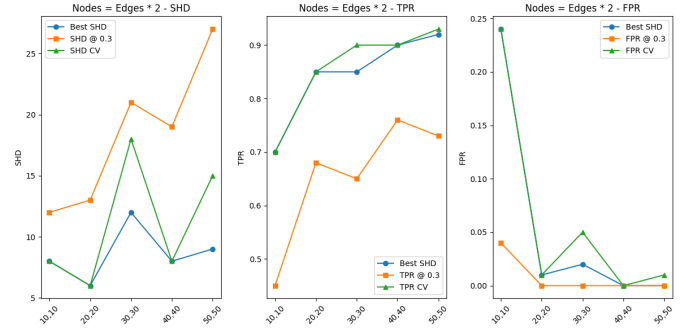


Fig. 8. Results for factor 2 configuration.

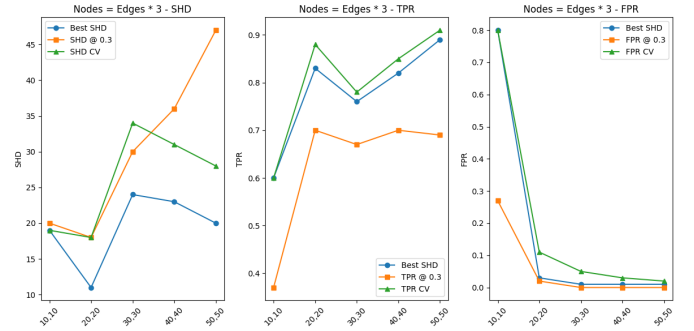


Fig. 9. Results for factor 3 configuration.